

DOM Events and Shadow DOM - Cheat Sheet

Event Capturing vs Event Bubbling

DOM events follow two main phases:

- **Capturing Phase** – Top-down: window → target
- **Bubbling Phase** – Bottom-up: target → window

```
// Capturing
parent.addEventListener('click', handler, { capture: true });

// Bubbling (default)
parent.addEventListener('click', handler);
```

Stopping Propagation

- `event.stopPropagation()`
 - Stops further propagation
- `event.stopImmediatePropagation()`
 - Stops all listeners
- `event.preventDefault()`
 - Prevents default browser action

Shadow DOM Basics

Shadow DOM provides encapsulation for styles and scripts within a component.

```
const shadowRoot = element.attachShadow({ mode: 'open' });
```

- **open**: accessible via
 - `element.shadowRoot`
- **closed**: inaccessible from outside

Shadow DOM Example

```
class MyElement extends HTMLElement {
  constructor() {
    super();
    const shadow = this.attachShadow({ mode: 'open' });
    shadow.innerHTML = `
      <style>p { color: red; }</style>
      <p>Hello from Shadow DOM</p>
    `;
  }
}
customElements.define('my-element', MyElement);
```

Event Bubbling in Shadow DOM

By default, events do not bubble out of a shadow tree.

```
const event = new CustomEvent('my-event', {
  bubbles: true,
  composed: true
});
this.dispatchEvent(event);
```

Quick Summary Table

Concept	Notes
Capturing	Top-down (less common)
Bubbling	Bottom-up (default)
Shadow DOM	Isolated DOM tree
<code>composed: true</code>	Event can escape Shadow DOM

Important Interview Questions

- ◆ What is the difference between event capturing and bubbling?
- ◆ What happens if both capturing and bubbling listeners are present on the same element?
- ◆ How can you stop an event from propagating?
- ◆ What is Shadow DOM and how is it different from Light DOM?
- ◆ What are the modes of Shadow DOM? What's the difference between *open* and *closed*?

- ◆ How does event bubbling behave inside Shadow DOM?
- ◆ What is the purpose of the *composed* flag in CustomEvent?
- ◆ Can you access the shadow root from outside the component? When and how?
- ◆ How do you style a component using Shadow DOM? Can global styles affect it?
- ◆ Give a real-world use case for Shadow DOM in modern web apps.